



CitySphere

NEBU Project Charter

CitySphere Smart-City Program

Project NEBU / CitySphere
Planning period September 2026 to July 2027

Under the supervision of Dr. Ghada

Handover control: This is an approval-stage planning baseline. Do not state intended benefits as achieved results before approved implementation baselines, comparison periods, and measured evidence exist.

Course Instructor: This charter was prepared under the supervision of Dr. Ghada.

1. Purpose, Objective, and Intended Impact

Purpose

Improve citizens' quality of life by using connected digital technologies to make city services more efficient, sustainable, accessible, and responsive.

Objective

Design, build, test, and operate a controlled NEBU implementation that connects traffic and mobility, waste management, energy and environmental monitoring, city operations, and citizen digital services. The implementation will validate the technical solution, operating workflows, security controls, citizen experience, and business case required for an evidence-led scale-up decision.

Intended Impact

The implementation is intended to provide earlier visibility of city conditions, clearer ownership of operational alerts, and more informed service responses. Intended outcomes include reduced congestion in selected corridors, fewer waste-overflow incidents, improved energy and environmental visibility, lower avoidable operating cost, faster issue response, and stronger citizen satisfaction. These outcomes remain unproven until approved baselines, targets, and evidence are available.

2. Scope and Constraints

In Scope

- Monitoring infrastructure and connectivity in approved selected districts.
- NEBU core platform, data layer, integration hub, dashboards, reporting, and KPI tracking.
- Smart traffic monitoring and congestion-alert support.
- Waste fill-level monitoring and collection-priority support.
- Energy and environmental monitoring.
- Citizen mobile services for core requests, alerts, city information, support, and feedback.
- City-management dashboard and accountable workflows: detect, assess, assign, intervene, verify, and improve.
- Cybersecurity, data protection, testing, training, implementation evaluation, and a scale-up recommendation.

Out of Scope

- Full citywide deployment.
- Legacy integrations not identified and approved during requirements.
- Hardware upgrades outside approved selected districts.
- Long-term, multi-year operations and maintenance contracts.
- Unvalidated AI prediction, automated recommendations, or autonomous operational decisions.
- Traffic-signal optimization unless specifically approved through requirements and change control; the defined baseline is monitoring and congestion-alert support.

Constraints and Open Decisions

Item	Status	Accountable owner	Required resolution
Selected districts	Open	Project Manager + City Authority	Approve during requirements.
City Authority Representative and approval route	Open	Project Manager	Confirm before M1 closes.
Device quantities, vendors, connectivity design	Open	CTO + COO + CFO	Approve before procurement.
Legacy-integration needs	Open	CTO + city operations	Confirm during requirements.
KPI baselines, targets, comparison periods, adoption target	Open	CFO + Project Manager	Approve before operation measurement.
Data-governance rules and AI boundary	Open	CTO + City Authority	Approve before relevant build/use.

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3. Budget and Financial Resources

The USD 5,000,000 ceiling is the total implementation planning budget, not a citywide rollout estimate. The primary source is sponsor and funding-partner investment approved through governance. The USD 300,000 contingency reserve is only for approved risk response or change; it is not discretionary spend.

Budget category	Amount (USD)	Share	Planned use
IoT sensors, smart devices, and field equipment	1,500,000	30%	Implementation monitoring infrastructure
Software platform, mobile app, and cloud infrastructure	1,000,000	20%	NEBU platform, dashboards, citizen services
Smart transportation and traffic systems	800,000	16%	Traffic monitoring and management support
Installation, networking, and infrastructure integration	600,000	12%	Field deployment, connectivity, integration
Project team, training, and technical support	500,000	10%	Delivery management, training, support
Cybersecurity, testing, and data protection	300,000	6%	Security controls, assurance, test activities
Contingency reserve	300,000	6%	Approved risk/change reserve
Total implementation budget	5,000,000	100%	Budget ceiling

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Controls: approve vendor scope, quantities, service levels, and acceptance criteria before purchase orders; report actual-versus-budget monthly; and require Steering Committee approval for contingency use or a material change.

4. Deliverables and Progress Controls

#	Deliverable	Acceptance / progress evidence	Accountable owner
1	NEBU platform and integration hub	Approved architecture, data interfaces, deployment record, integration tests	CTO
2	Traffic monitoring and congestion-alert capability	Field acceptance, alert-workflow test, dashboard evidence	CTO + COO
3	Waste monitoring and collection-priority support	Sensor calibration, data-quality report, workflow test	CTO + COO
4	Energy and environmental dashboard	Verified feeds, dashboard acceptance, exception handling	CTO
5	Citizen mobile application	Tested core journeys, accessibility/usability findings, support route	COO
6	City-management dashboard and KPI reporting	Approved KPI definitions, role-based access, operational drill record	COO + CFO
7	Security, privacy, training, and readiness pack	Security review, access-control evidence, training record, go-live checklist	CTO + COO
8	Implementation evaluation and scale-up recommendation	Baseline-to-target analysis, budget/schedule report, lessons learned, decision pack	CFO + Project Manager

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Progress is controlled through the approved schedule, delivery tracker, budget report, risk and issue registers, decision/change logs, test and acceptance records, availability/data-quality reports, and KPI scorecard. A phase cannot exit until its required evidence is accepted.

5. Timeframe and Decision Milestones

The dated baseline is September 2026 through July 2027, approximately 11 calendar months. Earlier material references a 12-month duration; the Steering Committee must approve one authoritative duration, phase-date baseline, and reporting basis before external commitment.

Milestone	Target date	Required outcome
M1 - Charter approval, planning, and requirements	31 October 2026	Charter, districts, stakeholders, requirements, workflows, KPI baseline/targets approved
M2 - Design and procurement	31 December 2026	Architecture, vendors, equipment, permits, connectivity, implementation plan approved
M3 - Platform and field implementation	31 May 2027	Platform, app, dashboards, sensors, integrations, operator preparation complete
M4 - Security and integration readiness	31 May 2027	Security, privacy, data-quality, integration acceptance complete
M5 - Testing and controlled go-live	June 2027	Go-live authorized; operation, issue response, and citizen support in place
M6 - Engagement, evaluation, and scale-up decision	31 July 2027	Onboarding/feedback complete; evaluation and scale-up decision submitted

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M1 must explicitly resolve the City Authority Representative, the traffic monitoring-versus-signal-optimization boundary, required legacy integrations, and whether any AI capability is allowed beyond future discovery.

6. Risks and Response Approach

Risks are reviewed weekly by the core team and monthly by the Steering Committee. Each risk must have an owner, preventive response, contingency action, target date, and status in the risk register. The full register is maintained in the NEBU Risk Management Plan (CSP-NEBU-06). The highest-priority risks are summarised below.

Risk	Likelihood	Impact	Owner	Preventive response	Contingency action
Cybersecurity or data-privacy incident	Medium	High	CTO	Encryption, least-privilege access, monitoring, audits, vendor due diligence	Isolate service; activate incident response; notify as required
System integration failure	Medium	High	CTO	Early interface testing, documented APIs, phased integration	Defer non-critical interface; use approved manual fallback
Network or connectivity failure	Medium	High	CTO	Redundant connectivity, monitoring, backup connections	Switch to backup network and offline data collection
Budget overrun	Medium	High	CFO	Phased procurement, monthly forecast, change control	Approved contingency or reduced non-critical scope
Low citizen adoption	Medium	High	COO	Accessible journeys, onboarding, support, feedback	Increase outreach and improve high-value features

Procurement or permit delay

Medium

Medium

COO / PM

Pre-qualified vendors, early authority engagement, delivery tracking

Alternate supplier or resequenced deployment

Poor data quality or sensor failure

Medium

Medium

CTO + COO

Acceptance tests, calibration, health monitoring, critical-zone redundancy

Replace device; flag data; temporary field inspection

Schedule delay

Medium

Medium

Project Manager

Weekly tracking, milestone reviews, resource adjustment

Reallocate resources and prioritise critical activities

Privacy or data-protection issue

Low

High

CTO

Privacy-by-design, restricted access, compliance reviews

Suspend affected service; conduct compliance review

Regulatory or approval delay

Low

Medium

PM

Early engagement, formal check-points

Escalate through governance and resequence work

7. Stakeholders and Governance

Stakeholder / role	Interest or responsibility	Named owner / engagement owner
Project Sponsor	Funding oversight and strategic decisions	Eslam Tarek (CitySphere)
Steering Committee	Approvals, governance, milestone decisions, material changes	City authority representatives, Sponsor, Project Manager
Project Manager & CEO	Planning, coordination, budget/schedule control, stakeholder communication, escalation	Mohammed Tarek
CTO	Architecture, software, integration, cybersecurity, technical acceptance	Eslam Tarek
COO	Field operations, workflows, training, citizen experience	Rahma Tarek
CFO	Budget, financial reporting, KPI analysis, evaluation	Nada Ahmed
City Authority Representative	Approvals, permits, regulatory liaison, acceptance	To be confirmed
Technology and field suppliers	Hardware, software, connectivity, installation, support	CTO + COO
Citizens and community representatives	Participation, adoption, accessibility, feedback	COO

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The Steering Committee meets monthly and at phase exits to review progress, budget, risks, readiness, and material changes. The Project Manager has day-to-day authority within the approved charter. Material scope, budget, schedule, or contingency changes require Steering Committee approval before implementation.

8. Team Roles and Responsibilities

Role	Named owner	Responsibilities
Project Manager & CEO	Mohammed Tarek	Overall planning, governance, coordination, budget/schedule oversight, stakeholder communication, escalation
Chief Technology Officer	Eslam Tarek	Platform architecture, software delivery, IoT and data integration, cybersecurity oversight, technical acceptance
Chief Operating Officer	Rahma Tarek	Operational implementation, UI/UX, service workflows, training, citizen experience, delivery coordination
Chief Financial Officer	Nada Ahmed	Budget control, financial reporting, data analysis, KPI tracking, business requirements, implementation evaluation
City Authority Representative	To be confirmed	City approvals, regulatory liaison, operational alignment, acceptance participation

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9. Success Measures

- Approved implementation scope delivered within the approved budget and authoritative schedule.
- At least 90% system availability during the operation.
- Measurable improvement in approved traffic, waste, energy/environmental, or citizen-service KPIs.
- Positive citizen feedback and achievement of the approved adoption target.
- No unresolved high-severity security, privacy, safety, or operational-control issue at the scale-up gate.
- A documented, affordable, operationally viable scale-up recommendation.

10. Approval

Approver	Role	Signature / date	Status
Eslam Tarek	Project Sponsor / CTO	Pending	Pending approval
To be confirmed	City Authority Representative	Pending	Pending appointment and approval
Mohammed Tarek	Project Manager & CEO	Pending	Pending approval

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Once signed, this charter is the baseline for scope, budget, schedule, roles, and decision authority. Changes must be recorded and approved through the governance process above.